Bridge test

Intro

On the opening day of the Millennium Bridge in London in the year 2000, hundreds of visitors crossing the bridge unintentionally induced lateral forces through their synchronized walking. This caused the bridge to begin wobbling sideways, a phenomenon that surprised engineers and shocked the public. As the oscillations increased, pedestrians adjusted their steps to maintain balance, which further amplified the movement. Due to safety concerns, the bridge was closed the following day to assess the situation and determine appropriate measures to ensure structural stability.

Test objective

The responsible engineers aimed to recreate the observed behavior under controlled conditions in order to precisely measure the bridge’s natural frequency and understand the cause of the oscillations. The goal was to identify the mechanism behind the synchronization of pedestrian movement and its interaction with the bridges dynamics.

Test setup

On the day after the closure, engineers organized a controlled experiment. Friends, family members, and colleagues were invited to participate. Groups of people were systematically led across the bridge in increasing numbers, while measurement equipment recorded vibrations and lateral movement.

Starting with 50 pedestrians the groups were led across the bridge for 20 minutes. Observing the results the groups were incresed by two to ten people and that process repeated.Up to 166 people, only slight and approximately linear oscillations were observed. However, beyond this threshold, the amplitude increased abruptly.

Results and conclusion

The bridge underwent a phase transition into large lateral oscillations due to synchronized pedestrian movement, and so the problem was recreated. As a solution, engineers installed damping systems and reduces coupling strings to reduce lateral motion.This successfully fixed the original issue and the bridge was safely reopened later.